

03 Case Study - AccountFlow OS

Source: docs/case-study/CASE_STUDY.md

Case Study - AccountFlow OS

Status: Day 5 complete

Live web: <https://accountflow-web.vercel.app>

Live API: <https://accountflow-api.vercel.app>

1. User & problem

Who: Alex Rivera, AE / client-delivery PM at a distributed B2B services firm.

Problem: After every client call she spends 30-40 minutes writing follow-ups, updating HubSpot, and filing Jira tasks - and still misses out-of-scope asks buried in the conversation.

Frequency: 8-12 calls/week (about 4-8 hours/week of post-call admin).

Business value: Cut post-call cycle time under 8 minutes while catching scope creep before it becomes unpaid work.

2. Existing workflow & bottleneck

See docs/workflow/CURRENT_STATE.md.

Typical path today:

1. Scramble for HubSpot notes and SOW PDF
2. Draft email from memory
3. Hand-edit CRM fields
4. Create Jira tickets one by one

Bottleneck: Manual glue between meeting memory, contract (SOW), HubSpot, Gmail, and Jira - with no verification layer. ChatGPT speeds drafting but still requires copy-paste and does not check the SOW.

3. Scope decisions & non-goals

In scope (v1): Upload or record transcript -> attach HubSpot deal -> attach SOW -> Scope Verifier -> human approve -> execute Gmail / HubSpot / Jira.

Out of scope (locked):

1. Auto-send without approval
2. Zoom/Meet bot that joins calls
3. Salesforce
4. Multi-tenant billing / org SaaS layer (open register + per-user BYOK instead)
5. Non-English transcripts

See docs/DECISIONS.md.

4. Architecture & major trade-offs

See docs/architecture/SYSTEM.md.

Choice | Why

Transcript-in (not meeting bot) | Privacy + 5-day feasibility

LangGraph + interrupt for approval | Explicit human gate before execute

FastAPI + Next.js | Clear API/UI split; deploy independently

BYOK vault (Fernet) | Operator does not hold shared HubSpot/Jira/LLM secrets

Turso in production | Local SQLite on Vercel wiped between cold starts

HubSpot pipeline stages via API | Stop guessing stage IDs in the approval UI

Key trade-off: We spend ~1 minute on approval to prevent wrong emails and CRM writes. That is intentional, not a bug.

5. Work delegated to AI vs judgment retained by humans

AI | Human

Transcribe (STT), extract actions, draft email/CRM/tasks | Approve or reject every section

Scope flags (IN/OUT/TIMELINE/BUDGET) | Override flags; choose change_request vs follow-up

Hallucination grader on weak transcripts | Edit low-confidence owners and values

Propose HubSpot stage / amount | Confirm against real pipeline dropdowns

6. Results

Measured sources only: docs/evaluation/BASELINE.md, docs/evaluation/RESULTS.md.

Metric	Before (manual)	After (AccountFlow)
Time to email + CRM (mean)	**38 min**	**6 min** (mock path)
Scope issues caught (baseline sample)	Y (if careful)	ChatGPT **N** **Y**
CRM fields correct (/5)	4.0	**4.5**
Automated suite	-	**12/12 PASS**

Primary target was <8 minutes - met on the timed mock path. Live LLM + real APIs will be slower; that must be re-measured in week 1 of adoption.

7. Failures, changes, limitations

Failures fixed (see docs/evaluation/FAILURES.md):

- * Short calls inventing deadlines (TC-06) -> grader + approval
- * Duplicate first names (TC-05) -> low confidence / TBD owner
- * Gmail 503 (TC-09) -> three retries, then clear error

Other product corrections during the sprint:

- * Session bounce on Vercel cold starts -> restore user from JWT; Turso for durable data
- * Cross-site cookies -> `SameSite=None` + `Secure` in production
- * HubSpot CRM fields as free text -> pipeline stage dropdowns; multi-deal create/select

Limitations (honest):

- * English-only v1
- * Scope Verifier quality tracks SOW quality
- * Owner assignment still needs human review when names collide
- * Per-call \$ cost and production P95 latency were **not instrumented**
- * Demo/eval often uses mock LLM/integrations; live keys change latency and failure modes

8. Two-week iteration plan

Week 1

- * Proxy user runs **5** anonymized real calls on the live URLs
- * Track: wall-clock time, edit count, execute success
- * Tune Scope Verifier false positives against real SOWs
- * Add simple per-run token/API cost logging

Week 2

- * Optional Zoom/Meet recording ingest **or** Slack notify on execute
- * Commitment tracker across runs for the same deal
- * HubSpot production OAuth review with security

Near-term backlog

- * Instrument P95 draft latency in production
- * Calendar-aware owner resolution
- * Stronger SOW parsing (structured sections, not only raw text)
- * Optional always-on API host if Vercel cold starts become painful for demos

9. Adoption metrics (planned post-deployment)

Metric	Week 1 target	Week 2 target
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Runs completed	5	10
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Avg time per run (live)	<10 min	<8 min
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Execute success rate	>90%	>95%
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Scope flag precision	measure	>85%
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